

Thematic Alignment Analysis of the Journal of Biomedical Informatics (2015–2024)

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Abstract

This study assesses the thematic alignment between articles published in the Journal of Biomedical Informatics (JBI) and the journal’s stated Aims & Scope over a ten-year period (2015–2024). A corpus of 2,039 abstracts was collected via the PubMed API. Each abstract was converted into a 384-dimensional semantic vector using Sentence-BERT, and cosine similarity was computed against the embedded Aims & Scope reference text. Results reveal a decline in mean alignment scores from 0.309 (2015) to 0.281 (2021), followed by a partial recovery, suggesting a period of thematic drift coinciding with the rise of AI and machine learning topics. Unsupervised clustering identifies two dominant research themes whose relative proportions show a notable shift over the study period.

Keywords: thematic alignment, biomedical informatics, sentence embeddings, cosine similarity, clustering, thematic drift

1 Introduction

The Journal of Biomedical Informatics (JBI) is a peer-reviewed journal publishing research at the intersection of medicine and computer science, indexed in PubMed and freely accessible. Over the past decade, the field of biomedical informatics has changed significantly, with artificial intelligence and machine learning becoming increasingly dominant topics.

When a journal’s published content gradually moves away from its stated editorial mission, this is referred to as *thematic drift*. This can happen when new research trends emerge faster than editorial policies are updated. In the case of JBI, it is worth asking whether the rise of AI-focused research has shifted the journal’s content away from its traditional scope.

This study measures the thematic alignment between JBI articles and the journal’s Aims & Scope over 2015–2024, using sentence embeddings and cosine similarity. We also apply unsupervised clustering to track how the journal’s thematic composition has changed over time.

1.1 Related Work

Sentence-BERT [1] introduced an efficient approach for computing semantic similarity between texts using siamese BERT networks, enabling meaningful comparison of scientific abstracts. BERTopic [2] combines embeddings with TF-IDF for interpretable topic modeling. Picascia et al. [3] applied similar techniques to study thematic structure across scientific corpora.

2 Research Question and Methodology

2.1 Research Question

The main question of this study is: to what extent have articles published in JBI drifted from the journal’s stated Aims & Scope over the period 2015–2024, and how has this alignment changed over time?

2.2 Formal Problem Definition

Let $D = \{d_i\}_{i=1}^N$ be the set of articles, where each d_i is represented by its title and abstract. Let s be the Aims & Scope reference text. Both are mapped to dense vectors using an embedding function $f(\cdot)$, producing $p_i = f(d_i)$ and $a = f(s)$. The alignment score for article i is:

$$\text{align}(i) = \cos(p_i, a) = \frac{p_i \cdot a}{\|p_i\| \|a\|} \quad (1)$$

Higher scores indicate stronger thematic alignment with the journal’s mission.

2.3 Pipeline

The methodology consists of four stages:

1. **Data Collection:** 2,039 abstracts collected from the PubMed API, covering the period 2015–2024. PubMed was selected as the data source because JBI is fully indexed there and the API provides free, structured access to abstracts.
2. **Embedding:** Texts converted to 384-dimensional vectors using Sentence-BERT (all-MiniLM-L6-v2). Sentence-BERT was chosen over simpler representations such as TF-IDF because it captures semantic meaning.
3. **Alignment Scoring:** Cosine similarity computed between each article embedding and the Aims & Scope reference vector. Cosine similarity was selected because it measures the angle between two vectors regardless of their magnitude, making it suitable for comparing semantic representations of texts with varying lengths.
4. **Clustering & Analysis:** K-Means clustering applied in the original 384-dimensional embedding space to identify dominant research themes. The optimal number of

clusters was determined using the Silhouette Score. UMAP was used solely for 2D visualization, as it preserves local structure better than PCA for high-dimensional data.

2.4 Embedding Model Selection

Two sentence embedding models were evaluated for this study. The first, **all-MiniLM-L6-v2**, is a general-purpose Sentence-BERT model that maps texts to a 384-dimensional vector space. The second, **S-PubMedBERT-MS-MARCO**, is a domain-specific model trained on biomedical literature, producing 768-dimensional vectors.

Both models were applied to the full corpus and the Aims & Scope reference text. The resulting alignment scores differed substantially. With **all-MiniLM-L6-v2**, scores ranged from 0.025 to 0.665 (mean = 0.295, std = 0.09), providing a wide distribution suitable for detecting thematic drift. With **S-PubMedBERT-MS-MARCO**, scores fell within a very narrow range of 0.845 to 0.946 (mean = 0.898, std = 0.02), offering insufficient discriminative power to identify drift or outliers.

Although PubMedBERT has a deeper understanding of biomedical language, it assigns very similar scores to all articles. A possible explanation is that the model was trained specifically on biomedical text, so it naturally places all biomedical documents close together in the embedding space. As a result, cosine similarity loses its ability to distinguish between more and less aligned articles.

For clustering, however, **S-PubMedBERT-MS-MARCO** produced more granular results: the Silhouette Score identified $k = 7$ as optimal, showing semantically coherent clusters such as Genomics & Cancer, Drug & Pharmacology, and COVID-19 — compared to only $k = 2$ with the general-purpose model. This shows that domain-specific models may be better suited for topic discovery, while general-purpose models are more appropriate for alignment scoring tasks. Since the main goal of our project is to see the thematic drift, **all-MiniLM-L6-v2** was selected as the primary embedding model for this study.

2.5 Ground Truth

The JBI Aims & Scope statement, retrieved from the journal’s official website, serves as the fixed thematic reference. Articles with a score above 0.4 are classified as highly aligned, corresponding to approximately one standard deviation above the mean ($\mu = 0.295$, $\sigma \approx 0.09$).

3 Experimental Results

3.1 Evaluation Metrics

Three metrics are used in this study. Cosine similarity measures the thematic overlap between each article embedding and the Aims & Scope reference vector, producing an alignment score between 0 and 1. The Silhouette Score is used to determine the optimal number of clusters. Finally, yearly mean, median, and standard deviation of alignment scores are computed to track thematic drift over years.

3.2 Dataset

A total of 2,039 article abstracts were collected from the Journal of Biomedical Informatics from the PubMed API, covering the period 2015–2024. JBI was selected because it has a clearly defined Aims & Scope statement and is fully indexed in PubMed, making data collection straightforward.

Data collection was performed using the Entrez API with the query "J Biomed Inform"[Journal] AND "2015/01/01"[dp]:"2024/12/31"[dp]. Articles without abstracts were excluded, as abstracts are the primary unit of analysis in this study. A small number of articles dated 2025 were also removed, as they likely reflect indexing delays rather than actual 2025 publications. The final corpus contains between 163 and 234 articles per year, with a mean abstract length of 244 words and a maximum of approximately 600 words.

The Aims & Scope statement was retrieved directly from the journal’s official Elsevier page and embedded using the same Sentence-BERT model as the article abstracts.

3.3 Alignment Score Distribution

The mean alignment score across the corpus is 0.295 (std = 0.09), with scores ranging from 0.025 to 0.665. The distribution approximates a normal curve centered around the mean, as shown in Figure 1. A total of 28 articles (1.4%) scored below 0.1 and were classified as outliers. The 10th percentile lies at 0.178, indicating that the lower tail of the distribution contains articles with notably weak thematic alignment.

The moderate mean score of 0.295 reflects the nature of the task: the Aims & Scope text is a short, high-level statement, while article abstracts cover specific methodological contributions. Perfect alignment is therefore neither expected nor necessary. What matters is the relative distribution of scores and how it changes over time.

We should keep in mind that cosine similarity scores in sentence embedding spaces are generally lower than we would think. Two semantically related texts may still produce scores in the 0.3–0.5 range, while completely unrelated texts tend to score below 0.1. This is consistent with our observed distribution, where a big part of articles cluster between 0.2 and 0.4, and only a small minority fall below 0.1 or above 0.5.

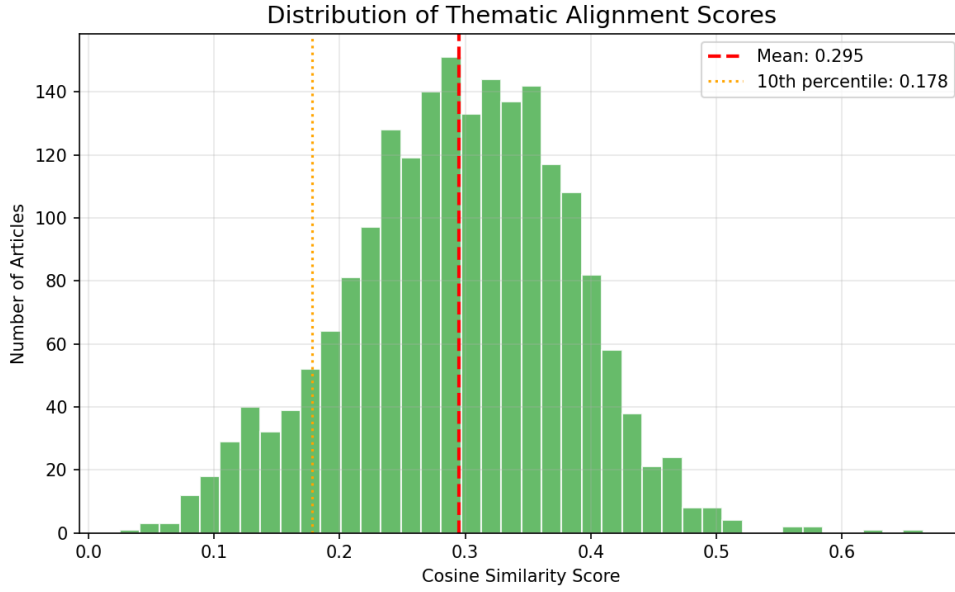


Fig. 1 Distribution of cosine similarity alignment scores across the 2,039 JBI articles. The dashed red line indicates the mean (0.295).

3.4 Thematic Drift Over Time

Figure 2 shows the mean alignment score per year. Scores declined from 0.309 in 2015 to 0.281 in 2021, then partially recovered to 0.301 in 2023. The proportion of highly aligned articles (score ≥ 0.4) dropped from 16.7% in 2015 to 5.8% in 2018, providing further evidence of thematic drift during this period.

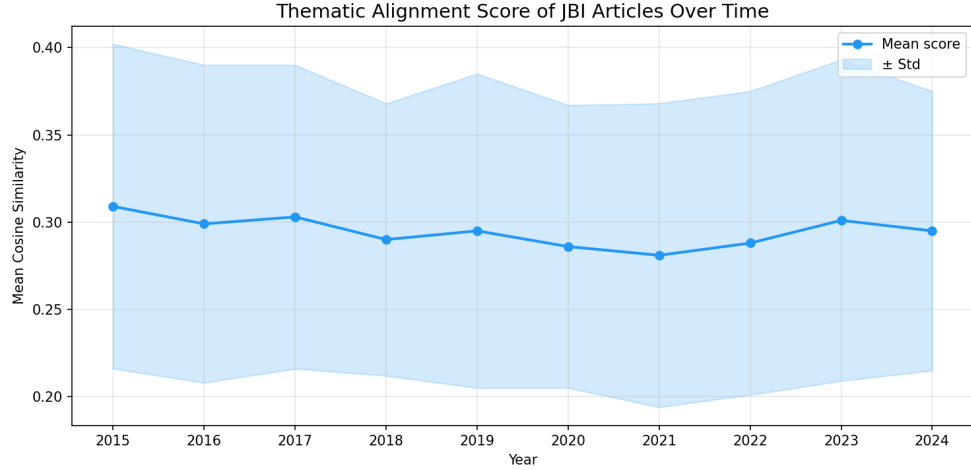


Fig. 2 Mean thematic alignment score of JBI articles per year (2015–2024). The shaded area represents $\pm$ one standard deviation.

3.5 Outlier Analysis

A total of 28 articles were identified as outliers (score < 0.1). Qualitative inspection reveals two patterns. First, a concentration of COVID-19 epidemiological modeling papers from 2019–2020, which focus on general statistical methods rather than biomedical informatics. Second, articles on general usability and interface design with limited informatics content. The lowest-scoring article (score: 0.025) addresses univariate time series forecasting with no explicit biomedical informatics context. Conversely, the highest-scoring article (score: 0.665), titled “Why is biomedical informatics hard? A fundamental framework”, directly addresses the theoretical foundations of the field, closely mirroring the journal’s stated mission. This qualitative validation confirms that the cosine similarity metric correctly captures thematic alignment.

3.6 Cluster Analysis

The Silhouette Score analysis identified $k = 2$ as the optimal number of clusters. Table 1 shows the cluster labels and top keywords.

Table 1 Cluster Semantic Signatures

ID	Label	Top Keywords
0	Clinical & Health Informatics	health, patient, research
1	Biomedical NLP & ML	learning, disease, medical

Figure 3 shows the UMAP projection of article embeddings colored by cluster. The proportion of Biomedical NLP & ML articles grew from 57.7% in 2015 to 74.2% in 2024,

while Clinical & Health Informatics declined correspondingly (Figure 4). This cluster consistently shows lower alignment scores, which explains the overall drift observed in the corpus. Outlier articles cluster predominantly at the periphery of the Clinical & Health Informatics group, suggesting that misaligned articles occupy the semantic boundary of the journal’s traditional scope rather than forming a distinct thematic group.

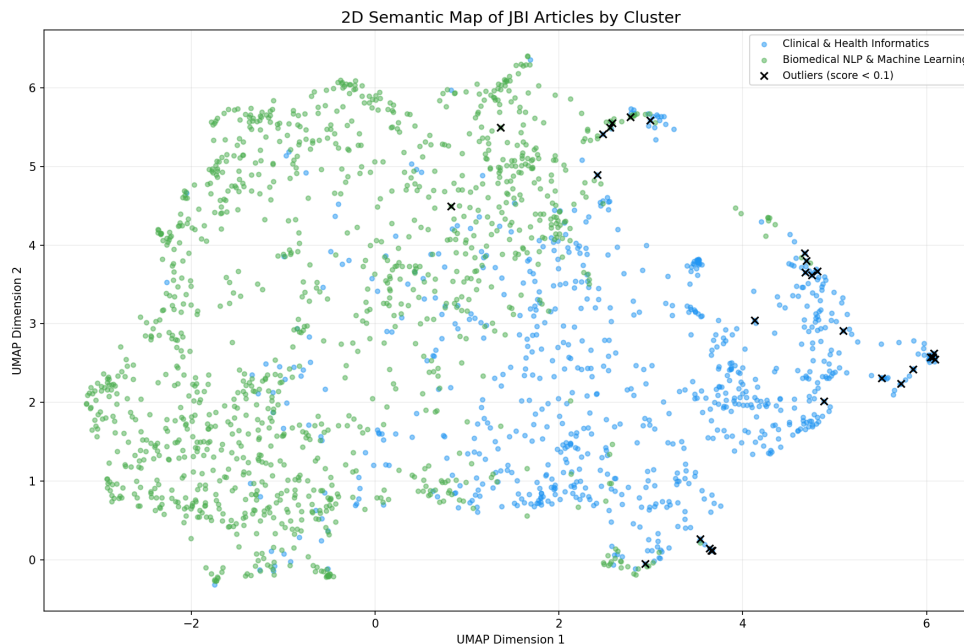


Fig. 3 2D UMAP projection of article embeddings colored by cluster. Black crosses indicate outlier articles (score < 0.1).

The Silhouette Score was checked for values of k from 2 to 10, and $k = 2$ produced the highest score (0.057). However, all values remained low, which is expected given that JBI articles are thematically similar by nature. It is worth noting that with the domain-specific PubMedBERT model, the optimal k was 7, producing more granular clusters. This suggests that the choice of embedding model directly affects clustering granularity, and that $k = 2$ should be understood as the most interpretable partition given our chosen model rather than an absolute ground truth.

The two identified clusters reflect a meaningful division of the field. Cluster 0 (Clinical & Health Informatics) groups articles focused on health systems, patient data, and clinical knowledge management — topics that closely mirror the journal’s traditional mission. Cluster 1 (Biomedical NLP & Machine Learning) captures the growing body of work applying computational and AI-driven methods to biomedical text and data, representing the newer direction of the field.

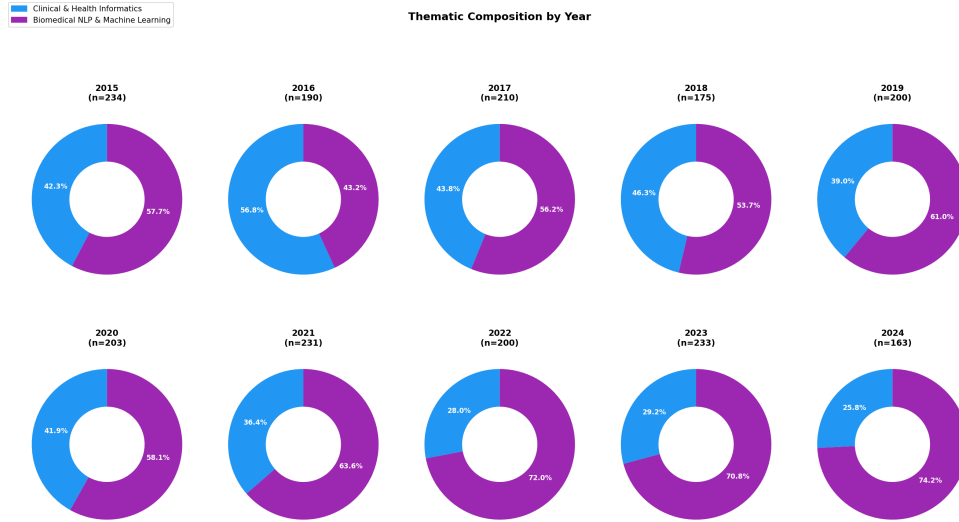


Fig. 4 Thematic composition of JBI articles by year.

3.7 Keyword Trends

The keyword trend analysis shows a clear rise in AI-related terms over the study period. For example, “deep learning” appeared in only 0.4% of articles in 2015 but grew to 16% by 2023. Similarly, “machine learning” increased from 10% to 21%. This is consistent with what we observed in the cluster analysis as these topics became more common, the share of Biomedical NLP & Machine Learning articles grew, and since this cluster has lower alignment scores on average, it helps explain the overall drift we measured.

4 Concluding Remarks

4.1 Conclusion

This study set out to measure thematic alignment between articles published in the Journal of Biomedical Informatics and the journal’s stated Aims & Scope over a ten-year period. The results show that thematic drift is real and measurable. Mean alignment scores declined from 0.309 in 2015 to 0.281 in 2021, before partially recovering to 0.301 in 2023.

The cluster analysis reveals that this drift is largely caused by the growing interest in Biomedical NLP & Machine Learning topics, whose share increased from 57.7% in 2015 to 74.2% in 2024. Since this cluster consistently shows lower alignment scores than the Clinical & Health Informatics cluster, its growth directly adds to the observed decline in overall alignment. This is further supported by the keyword trend analysis, which shows that “deep learning” grew from 0.4% to 16% and “machine learning” from 10% to 21% over the same period.

A notable finding is that many of the outlier articles come from 2019–2020, which overlaps with the COVID-19 pandemic. These articles tend to focus on epidemiological modeling and general statistical methods, rather than biomedical informatics. This suggests that the pandemic temporarily pulled the journal’s content away from its usual scope, which is something worth noting even if it was short-lived.

Overall, the results suggest that the rise of AI and machine learning has shifted the thematic composition of JBI over time. That said, the drift is not dramatic, the journal has not moved completely away from its original scope, but has expanded to cover newer trends.

4.2 Limitations and Future Work

Several limitations should be noted. First, the analysis relies on abstracts rather than full texts, which may not fully capture article content. Second, while a domain-specific model (S-PubMedBERT-MS-MARCO) was evaluated and found unsuitable for alignment scoring due to insufficient discriminative power, it produced more meaningful clusters than the general-purpose model. Future work could explore a hybrid approach, using domain-specific embeddings exclusively for clustering while retaining a general-purpose model for alignment scoring. Finally, the Silhouette Score suggested $k = 2$ as the best number of clusters, but the scores were low for all values of k we tested. This is likely because JBI articles are all quite similar to each other by nature, which makes it hard to find clearly separated clusters regardless of the method used.

There are several directions for future work. First, using full article texts instead of just abstracts would give a more complete picture of each paper’s content. Second, applying BERTopic could automate the topic labeling process and reduce the subjectivity in cluster naming since cluster labels were assigned based on the top TF-IDF keywords extracted from each cluster. Finally, testing other similarity metrics alongside cosine similarity could help improve drift detection, especially when using domain-specific embedding models.

References

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